COURSE OUTLINE

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Device Upgrade Automation with Claude

Space clear, firmware upload, pre and post checks and a scheduler, with a human on every reboot

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Enterprise IT and network operations teams, load balancer and data centre engineers, NOC automation leads, change and release managers in India and worldwide.

Prerequisites

- Basic TCP/IP and load balancer administration (F5, NetScaler or Radware)
- Comfort with a Linux shell, REST APIs and JSON
- Basic Python
- Familiarity with an AI coding assistant is helpful, not required

What you will be able to do

- Model a vendor-neutral device upgrade lifecycle: inventory, pre-checks, space clear, firmware upload, install, post-checks, rollback
- Set up Claude Code, the Claude Agent SDK and MCP servers that expose F5 iControl REST, NetScaler NITRO, Radware Alteon / Cyber Controller and Cisco APIC APIs with explicit scopes
- Build deterministic Python adapters that clear disk space, upload and verify firmware and install to the inactive boot location, HA-aware
- Diff pre and post snapshots and let Claude produce a structured verdict with rollback guidance
- Gate every consequential action behind ServiceNow change approval, schedule runs inside approved windows and keep a complete audit trail

Tools and technologies

Claude Code and Agent SDK | Model Context Protocol | Python | Pydantic

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F5 iControl REST and tmsh | Citrix NetScaler NITRO | Radware Alteon / Cyber Controller | Cisco APIC REST API | ServiceNow | GitHub Actions | pytest | Structured JSON outputs

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DAY 1 — 4 modules

MODULE 1

The device upgrade lifecycle and why upgrades fail

- Vendor upgrade mechanics side by side: F5 boot locations and UCS archives, NetScaler /var/nsinstall and build files, Radware Alteon image banks, Cisco ACI firmware groups
- The classic failure modes: full /shared or /var, corrupt image, HA pair upgraded in the wrong order, missing backup, change window overrun
- A vendor-neutral lifecycle: inventory, pre-check, space clear, upload, install, post-check, rollback, report
- The teaching pattern for the whole course: Claude plans and explains, deterministic code executes, a human approves, everything is logged

HANDS-ON LAB
Walk a real F5 and NetScaler upgrade runbook by hand in the lab, record every step and decision, and map each to the lifecycle stages.

DELIVERABLE
Annotated upgrade lifecycle map.

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MODULE 2

Claude Code, Agent SDK and MCP for upgrade workflows

- Installing Claude Code, project CLAUDE.md conventions and a Python Agent SDK scaffold for the upgrade runner
- Writing an MCP server that wraps F5 iControl REST (/mgmt/tm/sys/version, /mgmt/tm/sys/software), NetScaler NITRO (/nitro/v1/config) and Radware Alteon REST, or Cyber Controller for centrally managed Alteons, as read-only tools
- Tool use with strict JSON schemas: every tool returns typed, validated output, never free text
- Separating what the model may propose from what the code may do

HANDS-ON LAB
Build the read-only MCP server, connect it to Claude Code and ask Claude to describe the fleet using only tool results.

DELIVERABLE
Working MCP server and agent scaffold.

MODULE 3

Inventory and pre-flight discovery

- Pulling the device inventory and owners from the ServiceNow CMDB with the Table API (cmdb_ci_lb_* tables) and reconciling it with live device facts
- Collecting version, licence, platform, HA role, sync state and disk usage per vendor into one Pydantic inventory model
- Target version matrix and upgrade path rules (for example multi-hop F5 paths) as data, not prompt text
- Claude summarises readiness per device and flags inconsistencies between CMDB and reality

HANDS-ON LAB
Generate the readiness report for a mixed lab fleet and have Claude explain every device marked not ready.

DELIVERABLE
Fleet readiness report.

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MODULE 4

Pre-checks and the golden snapshot

- Virtual server, pool member and monitor status, HA failover and config-sync state, certificate expiry, licence validity
- Configuration backups before any change: F5 UCS via tmsh save sys ucs, NetScaler system backup via NITRO, Alteon configuration export
- A versioned JSON snapshot schema that the post-check module will diff against
- Hard stop rules: an upgrade does not proceed when any pre-check is red or a backup is missing

HANDS-ON LAB
Implement the pre-check and snapshot module with pytest coverage, then break a backup on purpose and watch the run stop.

DELIVERABLE
Pre-check gate and golden snapshot.

DAY 2 4 modules

MODULE 5

Disk space clearing and firmware staging

- Safe space reclamation per vendor: old F5 software images and UCS archives, NetScaler /var/nsinstall, /var/core and /var/crash, stale Alteon images
- Allowlists of deletable paths and a dry-run mode that reports what would be removed and how much space it frees
- Firmware upload with iControl REST software-image uploads and NITRO systemfile, with SHA256 verification against the vendor checksum
- Claude drafts the space-clear plan from disk facts; the adapter executes only allowlisted deletions after approval

HANDS-ON LAB
Fill a lab device disk, run the dry-run plan, approve it and stage the firmware image with checksum verification.

DELIVERABLE
Space-clear and staging adapter.

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MODULE 6

Upgrade execution adapters, HA-aware

- F5: install to the inactive boot volume via iControl REST or tmsh install sys software image, then reboot into it; NetScaler: NITRO install and reboot; Alteon: image bank update and reload
- HA ordering: upgrade standby, verify, fail over, upgrade the former active, confirm sync
- Idempotent, resumable steps with state recorded after every action so a failed run can continue or roll back
- Claude proposes the ordered plan and the exact API calls as structured JSON; the runner executes the approved plan only

HANDS-ON LAB

Upgrade an HA pair of lab load balancers end to end through the runner, with a mocked failure halfway to test resumption.

DELIVERABLE

HA-aware upgrade runner.

MODULE 7

Post-checks, verification and rollback

- Capturing the post-upgrade snapshot and diffing it against the golden snapshot: version, virtual server status, pool members, HA sync, certificates
- Claude reads the diff and returns a structured verdict (pass, warn, fail) with a plain-English explanation for the change ticket
- Rollback paths: boot the previous F5 volume, NetScaler previous build, Alteon previous image bank, restore from backup
- Evidence pack generation for the change record

HANDS-ON LAB

Run the post-check diff on a successful and an intentionally broken upgrade, then execute the rollback path for the broken one.

DELIVERABLE

Verification diff and rollback module.

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MODULE 8

Guardrails, approvals and audit for upgrades

- Allowlisted tools and devices per run, read-only by default, destructive actions behind explicit approval
- ServiceNow change requests created through the Table API and approved through Flow Designer before space clear, install and reboot
- An append-only audit log of every prompt, plan, approval and API call, with secrets kept out of prompts and logs
- Testing the guardrails with pytest: prompt the agent to skip a check and prove the runner refuses

HANDS-ON LAB
Wire the approval gate to a lab ServiceNow instance and attempt an unapproved reboot; inspect the audit trail afterwards.

DELIVERABLE
Approval gate and audit log.

DAY 3 3 modules

MODULE 9

Scheduler and change windows

- Reading approved change windows from ServiceNow and refusing to start outside them
- Scheduling runs with GitHub Actions schedule and workflow_dispatch, or an in-house APScheduler service, with a persistent job state store
- Pre-window reminders, in-window progress and post-window summaries to the operations channel
- Handling overruns: pause at the next safe step, request a window extension, never leave a device mid-upgrade

HANDS-ON LAB
Schedule a batch upgrade in the lab, let it start automatically inside the window and watch it pause when the window is about to close.

DELIVERABLE
Scheduled upgrade pipeline.

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MODULE 10

Cisco ACI and fleet-scale, multi-vendor runs

- Cisco ACI upgrades as Cisco prescribes: the ACI pre-upgrade validator, APIC firmware repository and firmware groups via the APIC REST API, APIC cluster first then switches in groups
- Batch strategies for large fleets: blast radius limits, canary devices and stop-on-first-failure
- Extending the adapter pattern to other load balancers (Anaya LB) with the same lifecycle contract
- Reporting in GitHub pull requests: plan, evidence pack and audit summary reviewed before the next batch

HANDS-ON LAB
Run the ACI pre-upgrade validator and build an APIC firmware group plan, then execute a canary-first batch across the mixed lab fleet.

DELIVERABLE
Fleet batch runner and ACI plan.

MODULE 11

Capstone: scheduled, approved, HA-aware upgrade end to end

- Inventory in, verified upgrade out: pre-checks, space clear, upload, install, post-checks and rollback readiness on a mixed fleet
- Wire the scheduler, approval gate, HA-aware runner and post-check diff into one pipeline driven by a device list and target version matrix
- Evidence pack and audit summary delivered as a GitHub pull request for review
- Acceptance criteria review, limitations statement and live demonstration to the group

HANDS-ON LAB
Take a mixed lab fleet from device list to evidence pack through the full pipeline, then demo the run, the approvals and the audit trail to the group.

DELIVERABLE
End-to-end upgrade automation and runbook.

Capstone

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Capstone project

A repeatable upgrade workflow that takes a device list and target versions, runs pre-checks, clears space, stages verified firmware, upgrades HA pairs in the right order inside an approved change window, verifies the result against a golden snapshot and ships an evidence pack, with no reboot ever executed without a human approval.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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